Adjacency, Aggregation, and Proration Challenges

Overview:

The goal of these exercises is to provide you with an opportunity to try out some of the steps in the redistricting data process workflow. There definitely won't be enough time during the class meeting today to completely solve these challenges but a) I hope that they help you build intuition for why problems in this field can be difficult and b) if you get interested in working on them I hope you'll continue to try to come up with solutions.

Warning: These challenges are fairly representative of real-world data processing and are unfair in many different ways!

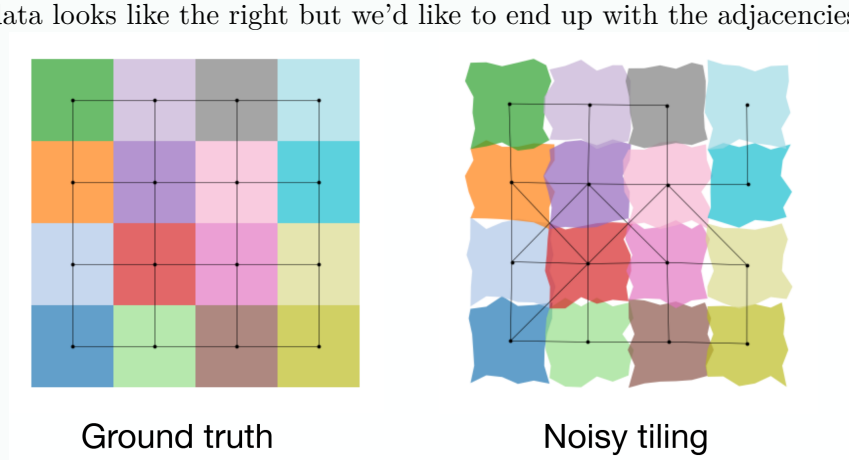
Data

You can download the datasets in a single .zip folder from https://tinyurl.com/GEOG224DATA. There are five shapefiles in the .zip folder:

- noisy_toy_grid_crs.shp contains a model of a badly constructed precinct map for problem 1 below
- blocks.shp is the census blocks with population and demographic data for Providence, RI
- precincts.shp contains the boundaries of the 2018 precincts in Providence with some voting data from 2016 and 2018 elections
- new_precincts.shp contains the boundaries of the 2020 precincts in Providence with no tabular data
- districts.shp contains the ward boundaries for Providence's city council

Problem 1: Noisy Adjacency

The given data looks like the right but we'd like to end up with the adjacencies on the left!



Your challenge is to use the tools in ARCGIS to process the shapes so that you can recover the "correct" adjacencies. Ideally in as automated a way as possible so that if I generated a new noisy tiling with a different random seed, you could still recover the correct graph.

Problem 2: Precincts in Providence

In order to study the properties of the ward boundaries in Providence, RI, we need to get a single dataset together that includes the current precinct boundaries (new_precincts), the population and demographics of each precinct (blocks), which ward each precinct belongs to (districts), and how many people voted for each candidate in each precinct (precincts). Unfortunately, this data is scattered across a bunch of different units and you'll need to assign, aggregate, and prorate the data to get a working dataset.

In order to break this challenge into tractable subtasks, the questions below walk you through some steps on individual data units before prompting you to try the same analysis on the entire file.

Single Units:

1. Which Ward (district) is precinct "Providence 63" from the new_precincts file inside?
2. How many people live in precinct "Providence 7" from the new_precincts file.
3. In block 5001, how many people voted for the Democratic Gubenatorial candidate in 2018?
4. In precinct "Providence 60" how many people voted for the Democratic Gubenatorial candidate in 2018?

Full Data Process:

5. Add a new column to the new_precincts file where each precinct is labelled with the ward it falls inside.
6. Add a new column to the new_precincts file where each precinct is labelled with its population by aggregating the values from the blocks.
7. Add new columns to the new_precincts file where each precinct is labelled with the vote totals and registered voter numbers from the elections on the precincts file.

Answering Questions:

8. For each of the new_precincts compute the proportion of voters that voted for the Democratic Presidential candidate in 2016.
9. Using the ward assignments, compute the population imbalance of the plan as a whole.
10. Construct a new districting plan by changing the assignments of the wards so that your new plan is at least as balanced as the enacted one but the partisan outcome is more (un)fair than the enacted one.